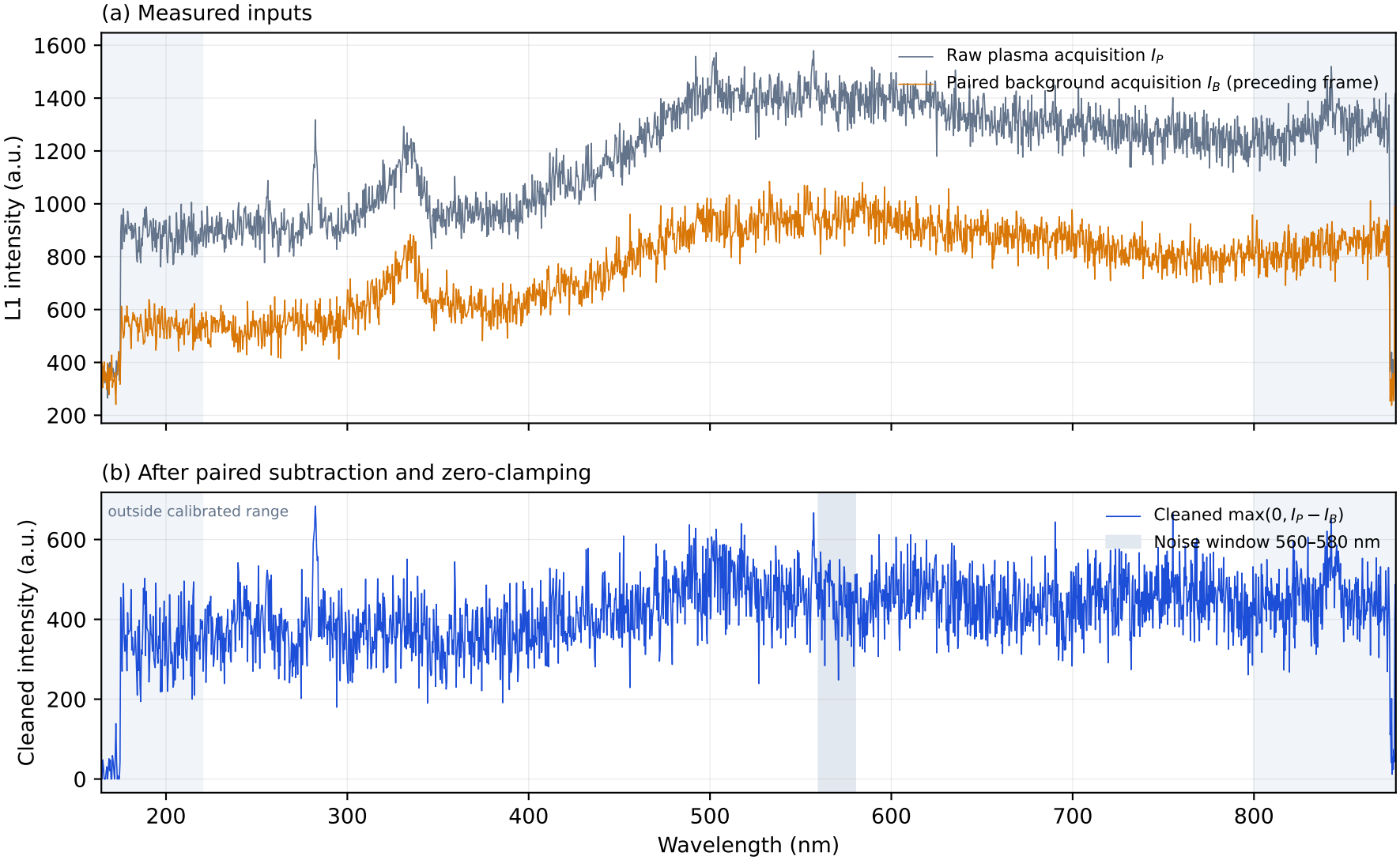

(a) Measured inputs
L1 intensity (a.u.)
1600
1400
1200
1000
800
600
400
200
Raw plasma acquisition I_P
Paired background acquisition I_B (preceding frame)
(b) After paired subtraction and zero-clamping
outside calibrated range
Cleaned intensity (a.u.)
600
400
200
0
Cleaned max$(0, I_P - I_B)$
Noise window 560–580 nm
200
300
400
500
600
700
800
Wavelength (nm)